

TRIANGLE

MESIER 1.6

— TRIANGLE MESHER —

Spatially-Variably Refined Meshes from PSLG's

Constrained-conforming 2D Delauney triangulations can be created from "parallel straight line graphs" with the very efficient and flexible program **Triangle** (J. Shewchuk, <http://www.cs.cmu.edu/~quake/triangle.html>). Please refer to the documentation of this program for details on how to operate it (on the command line type "triangleMesher -h" in the directory where the triangleMesher.exe lives). The utility program **GeometryPreprocessor.exe** by SKM is provided with CSP to create ".poly" input files for Triangle using ".dxf" files as input (DXF is a graphics interchange format created by AutoCAD. One must make sure that the DXF file format is compatible with the original DXF standard specification (new AutoCAD files are not, those up to version 12 are). A number of other rules must be adhered to when creating a DXF file for the *GeometryPreprocessor* program:

1. The first polygon in the drawing must be the bounding rectangle/graph (drawn as a closed polygon, not as a rectangle and insuring that the origin is in 0.0, 0.0).
2. The absolute dimensions of the drawing will be those of the final model. Change the drawing scales accordingly.
3. Draw polygons and lines, but do not smooth any of these. Also duplicate polygon segments must be avoided. Thus, at least one of two polygons which share a boundary, must be an open polygon.
4. "Triangle" expects minimum angle bounds for the triangular finite-elements it generates. These should not be violated by any intersections of lines in the drawing or by angles inside the polygons themselves (Triangle cannot discriminate between wanted and unwanted angle bounds and will therefore continue to refine the model until it runs out of numerical precision). The angle bound (option -qxx.) can be used to influence element size gradients in the mesh. The larger the bound the more smoothly transitions will be.
5. Triangle will need "material points" to assign properties to polygon- or line-delimited regions in the model. One can pick such points manually and create a regional property list inside of the ".poly" file (see instructions in Triangle documentation for the format of this list). Alternatively, (Option 2 in G.P. menu) one can let the *GeometryPreprocessor* assign these points on the basis of specific fill colors applied to closed polygons in the drawing. The *GeometryPreprocessor* program will prompt the user for property values which shall be assigned for each color. However, preparing the material points list by hand (Option 1) is highly recommended over the more advanced option 2 since the conversion algorithm is much more robust.
6. One can also use drawing layers in DXF files to discriminate among objects which should or should not become part of the mesh. The *GeometryPreprocessor* expects at least one drawing layer specification. If the drawing program does not indicate the name of the default layer, the DXF file can be examined directly since it is in text format. Many drawing programs call the initial layer DEFAULT layer. Use input layer names in capital letters since DXF file content is not case sensitive.

PLANAR

NO SURFACES!

606/89/79
759/51/131
2590.29x
0.25x
-0.18 ✓
16-0.16 ✓
17x 0.0444
0.0222

7. In the ".poly" files one can specify a maximum triangle size for each PSLG region. Otherwise, Triangle will make the mesh as coarse as possible given the amount of geometric detail in the model. The *GeometryPreprocessor's* Option 2 will assign default size bounds which can be edited before the meshing.
8. The PSLG must be exact ! - Duplicate region boundaries must be avoided and filled polygons which overlap each other cannot be used to mimic cross-cutting relationships. Triangle removes duplicate points, but the more of such glitches occur the more likely it is that one will get a corrupted mesh. The error reporting of Triangle (options -cv), is very good.
9. When running Triangle (on a PC or Unix/Irix machine) one must give angle bounds for the triangulation and also request a "*.neigh" file needed by CSP to establish neighbors of each finite element. The suggested set of command line arguments for Triangle is (run `triangle -h` for what the options mean):

```
triangle -a -A -e -p -q30. -n -O -V myfile.poly
```

The *GeometryPreprocessor* (Option 2) program helps to identify faulty input geometries by writing a separate DXF file (*.faulty.dxf) containing polygons that violate the angle bounds for Triangle. This, however, is useful only for input geometries which consist entirely of closed polygons. Furthermore, Option 2 reports if it cannot find a unique material point for a polygon by outputting the coordinates of the problematic polygon as well as those of the polygon which contains the ambiguous material point. In most cases the reason for this type of error are duplicate or overlapping polygons. The *GeometryPreprocessor* (Option 2) will fail to detect small angles that arise where polygons intersect. Such locations (in model coordinates) will however be reported by 'Triangle' which is useful for fixing these errors.

Triangle creates ".node, .ele" and ".neigh" files with a number corresponding to the refinement level (one can refine the same mesh several times to test for mesh convergence and mesh sensitivity of a computation). With the verbose meshing option, some useful mesh diagnostics will be reported in the meshing process. One can also examine the quality of meshes visually, with the program "ShowMe". This program was also written by J. Shewchuk but runs only on platforms with X-windows capabilities.

While a Triangle output mesh may be perfectly fine in terms of angle bounds etc., it may still contain knots where clusters of extremely small elements were created. This is most certainly the case when the reported mesh statistics indicate the existence of elements with an area below $1.0e-3$. In this case two very closely spaced points in a polygon, or a failed point snap leaving a minute gap in the model, are the most likely culprits.

"Triangle"-generated meshes are imported into CSP using its mesh interface which is encapsulated into the class **MeshInterface**. The code for this is simple:

```
CSP_TRIANGLE_Interface  mesh_interface;
CSP_VSet<csp_float,2>   vset;
char                    fname[200];
```

```
cout << "\nmain: Enter name of 'Triangle' input file set: ";
cin >> fname;
mesh_interface.ReadTriangle2DMesh( fname, vset );

SuperGroup<csp_float,2> my_new_model( vset );
```

The method `ReadTriangle2DMesh()` reads the set of input text files (.node, .ele, .neigh) created by Triangle and puts the results into the `CSP_Vset` with the name 'vset'. Property values will be stored and interpreted as the element variable "permeability". A fast way of arriving from this data at desired values of other properties is to first create element Group objects for each of the permeability values and then use the Group class interface to modify the corresponding regional properties:

```
set<string> group_names;
my_new_model.GroupsFromPropertyValues( "permeability", group_names );
```

In this example, the names of the new groups are output to an STL set of group names.

Trouble Shooting. There are many checks that CSP performs on mesh topology and additional checks can be invoked while running Triangle by using the `-C` option. If one still gets a message from the Solver, e.g., `DIAGONAL NOT POSITIVE`, the most likely reason for this is that one of the region points failed to lie inside the target region and the Triangle flag `-O` (no holes) was specified. In this case, Triangle assigns a zero material property value to the Elements in that region. Clearly this value will be propagated into the global solution matrix. One can see whether this is the case by examining the `*.1.ele` file with the material properties of the elements.

PSLG CREATION IN RHINOCEROS

The following documentation stems from Mandefro Belayneh's PhD thesis entitled "Analysis of natural fracture networks in massive and well-bedded carbonates and the impacts of these networks on fluid flow in dual porosity modeling." (Imperial College London). Mandefro created extremely intricate fractured limestone models with Rhino and encountered all possible caveats of PSLG's in this process.

CAD representation: as points, lines, poly-lines in 2D, surfaces and poly-surfaces in 3D.

Initial step: Fracture trace maps representing the geometry of fractures saved as Adobe Illustrator as *my file.ai* or *my file.dxf* format can be used as initial step. These are then imported to the Rhinoceros2.0™ commercial software package. In the 'properties' dialogue box, initial settings are set before proceeding to the next step. Unless otherwise stated, all the measurements are in SI units.

Set the following properties: **Absolute tolerance** (0.001) in metres is one order of magnitude smaller than the smallest feature in the model. The smallest feature in the model is 1mm by default. **Relative tolerance** (0.01) is the ratio of the smallest to the largest feature expressed as percentage. If the smallest and largest features in the model have magnitudes of 0.001m and 10m, the relative tolerance will be 0.01. **Angle tolerance** is by definition 0.01 and determines the quality of mesh.

The *grid spacing* dialogue box shows *snap spacing*, *grid spacing*, *major lines every* and *grid extents*. These properties are dependent on the size of the 2D map under consideration. After setting these properties, the joint/fracture trace map can be imported to Rhinoceros.

To assign an aperture for each joint trace map, each joint has to be offset by the magnitude of the effective hydraulic aperture of the joint (e.g. 0.003 and 0.005m). Joint aperture is the distance measured perpendicular to the plane between the opposite walls of a joint. Joint surfaces are far from being planar and the distance between the two varies along the fracture. The boundary object that bounds the fracture trace map on the **default layer** is carefully redrawn. The fracture traces are drawn as poly-lines. The poly-lines in the middle of the model are closed poly-lines whereas the ones adjoining the boundary object are open. All points along the boundary lines intersecting the boundary object should be precisely snapped.

On a different drawing layer, each region has to be defined by material points. Each fracture bounded block is bounded by a poly-line defined by material points with co-ordinates (x,y) in the 2D plane and (x,y,z) in 3D space. These points serve to uniquely identify fracture bounded blocks (matrix) and fractures within the model, as shown in Table 1. The assignment of the material points shown in Table 1 can be done using Microsoft Excel and saved as a tab-delimited text document.

Table 1 Material points assigned to joints and joint bounded blocks (matrix).

1	x ₁	y ₁	z ₁	k ₁	A ₁ (all columns are tab-delimited)
2	x ₂	y ₂	z ₂	k ₂	A ₂
.	.	.	.	.	.
.	.	.	.	.	.
n	x _n	y _n	z _n	k _n	A _n

where n represents the number of the material point within the model. The first column is a positive integer ranging from 1 to n allocated to each data point, the second to the fourth column represent the position in the 2D plane or 3D space, the fifth column represents permeability and the last column is the maximum area of the largest finite element allowed in the mesh of this region of the finite element model. Because of their different sizes, fractures and matrix will have different mesh sizes. This adaptation to size gives a unique possibility of representing and analysing the smallest features in the model. In this model, joints and matrix have permeability (k) values of $1.00\text{e-}12\text{m}^2$ and $1.00\text{e-}15\text{m}^2$ respectively. If the joints are interconnected, a single value of permeability will define the property of the network. To distinguish each block so that it can later be addressed individually, permeability variation of $1.00\text{e-}10\text{m}^2$ has been added in order to simulate a realistic variation of permeability that would be found in rocks. This artificially enhances the permeability and enables the variations encountered in naturally fractured rock masses to be represented. In this work the specified apertures are large enough to ensure that joint permeability is always greater than the permeability of the matrix unless otherwise stated. The complete view of the joint trace map drawn using Rhinoceros commercial software with the material points assigned is shown in Fig. 1.

Fig. 1 to be shown here if necessary

The default layer is saved in 'Rhinceros' as **my file.3dm** and **my file.dxf** in AutoCad release 12. The dialogue box appears and the settings **entities only**, **polyline maximum angle (0.01)**, **polyline chord height (0.001)** and **polyline segment length** are selected. The layer containing the material points will be saved as **my file.csv** and then the extension converted to **my file.txt**.

Run a GeometryPreprocessor .exe

This program is an input file for triangleMesher and helps to identify faulty input geometries such as those violating the angle limit of 30°. To run this command, type 1 and press enter. **Main: enter the name of the dxf file:** type the file name without the extension and press enter. **Please enter the number of the drawing layers to be read from file:** type 1 and press enter. **Please enter the name of the desired layer:** type **DEFAULT** all in upper case letters. This program reads polygon(s) and lines from **AutoCad.dxf** format and converts polylines to triangleMesher input such as **my file.poly** and **my file.vtk** files. Open **my file.poly** and type the number of material points described in Table 1 and type 1 separated by a tab. Then copy and paste the five columns from Microsoft Excel saved as tab delimited ***.txt** format from Table 1 in **my file.poly** created by the GeometricPreprocessor.exe program.

triangleMesher (command prompt window)

Go to (change directory) the directory where the triangle mesher is located and type **csp-etc/meshing/triangleMesher -aAnepCVq30. my file.poly**. The combination of letters signifies a range of options available through -help. The number 30 denotes the minimum angle criterion in the triangle mesh (i.e. that the angle between two poly-lines within the mesh should be greater than or equal to 30). The mesher creates five output files with the extensions ***.1.poly**, ***.1.ele**, ***.1.node**, ***.1.neigh** and ***.1.edge**. The maximum number of points, triangles, segments, encroached segments and bad triangles will be displayed on the screen. In the example shown (Fig. 2) 623,975 nodes, 1,897,141 segment faces and 1,264,167 triangles are allocated. In addition to this information the mesher shows the minimum area of individual triangles in the mesh which should be greater than 1.00e-06 in this analysis.

Run CSP3D_Mesh_to_VTK.exe

Type **my file.1** without extension. In this analysis the main element width range, element area range and model dimensions (x,y) 0-x and 0-y in metres will be displayed on the screen. The program checks the existing files and creates the files **mesh0.vtk** and **log10-permeability-node.vtk** which will later be displayed in the viewing directory. It also shows **SMALL_ELEMENTS-mesh.vtk** when the areas of the finite elements are less than or equal to 1.00e-06 and **ZERO_ELEMENTS_mesh.vtk** when there are regions in which material points are absent. Small elements will be created if extremely small regions (less than the value in the default) accidentally exist in the model or when the poly-lines (traces of joints) are not properly snapped to boundaries in 'Rhinceros'. Before proceeding to the next step, regions with no material point should be assigned one, and small areas need to be removed. The mesh can be displayed by copying and

RHINO:

EXPORT
TO
DXF

SO GROSS
WIKED
MODEL

→ Generates file.poly and file.vtk

confusing...

RHINO?

DO NOT GPUT LINES WITH LINES.
SPLIT USING POINTS!

COMPILING TRIANGLE

(for windows XP)

makefile:

-D NO_TIMER

-D CPU86

-D REDUCED : disables research
interest features.

-i -F -s -C

pasting **mesh0.vtk** and **log10-permeability-node.vtk** files into the VTK examples directory as shown in Fig. 2.

Compared to the 'finite grid difference methods' in which all objects are discretized with equal resolution, the unstructured grids of the combined finite element and finite volume methods have the advantage of flexibility in grid size and are therefore capable of representing the smallest features, such as fractures, reasonably well (Fig. 2). In finite difference methods, the properties of the material are assumed to be continuous and uniform. A continuous material within the model can be represented as a set of discrete points arranged as an equally spaced grid. The unknown variables in the equation are solved at the nodes. Associated with each node or block are the known values. There are two types of nodal representations: (i) 'block centred grid' where the nodes are placed at the centre of the grid and (ii) 'mesh centred grid' where the nodes are placed at the intersection of the grids. The finite difference method is commonly employed in hydrogeology where mesh- and grid-centred methods are most convenient in situations where head and flux are specified respectively (Fetter, 2001). The grid is regular with the rows and columns representing small increments in x (i.e. Δx) and y (i.e. Δy) directions respectively from the reference point.

Figure 2 be shown here if required

Figure 3 to be shown here if necessary

Figure 4 can be included here

Copy and paste both **mesh0.vtk** and **log10-permeability-node.vtk** created by running **CSP3D_Mesh-to_VTK.exe** and **my file.vtk** created when running the **GeometricPreprocessor.exe** program into **PermeabilityViewer**. This will show conductive areas in red and the surrounding matrix in blue (Fig. 4). Thus it can be used to detect incorrectly assigned areas, for example when the joint traces are not properly snapped to the boundary line or matrix blocks are not constructed from closed poly lines. In the latter case, the model will flood with red indicating that the matrix has the same permeability as the fractures.

CSP3D_linear_flow_analysis.exe

Copy and paste the five files ***.1.poly**, ***.1.ele**, ***.1.node**, ***.1.neigh** and ***.1.edge** created by running the **triangleMesher.exe** program in the working directory and write the configuration file as shown in Table 2.

Table 2 Configuration file for the linear flow analysis in CSP3D.

'my file.1' model configuration file for CSP3D example file set

joints	permeability	1.0e-12	1.0e-12
matrix	permeability	1.0e-15	1.0e-15
# material properties applied to the whole model			
porosity			0.25
storativity			1.0e-9
fluid volume source			0.0
fluid pressure			0.0

property assignment to model subregions

joints	complete	porosity	1.0
joints	complete	permeability	1.0e-9

global essential conditions (in this case boundary conditions)

LEFT	Dirichlet	fluid pressure	1.0e+7	1.0e+7
RIGHT	Dirichlet	fluid pressure	0.0e+0	0.0e+0

CSP3D_linear_flow_analysis.exe reads *.1.node, *.1.ele, *.1.neigh and this configuration file displays the properties of the model described in Table 2. The permeability at each nodal point, each finite element and neighbouring elements will be computed. The user will then be prompted to indicate whether he/she (i) requires the output in the form of JPEG images of the pressure and velocity distribution or (ii) if the user would like to group integral by group volume or if he/she (iii) would like to vary permeability in a single group or if he/she (iv) would like to model transient flow.

The output from the linear flow analysis is a series of files for the visualisation toolkit (VTK, Kitware.com) such as **fluid-pressure0.vtk**, **nvelocity0.vtk**, **nvolume-flux0.vtk**, **volume-flux0.vtk**, **velocity0.vtk**, CSP-monitored regions in text format, and a flow velocity histogram for the model. **Fluid-pressure0.vtk** depicts the pressure distribution within the model for a given far-field pressure. The output file **velocity** shows the fluid flow velocity in the finite elements and **nvolume flux** shows the nodal volume flux within the model. Flux is extrapolated to the nodes so that it can be visualised using rainbow colour shading. **Flux_histogram.bins** creates **joints-volume_flux.txt** and **matrix-volume_flux.txt** which are the velocities of fluid within the joints and matrix respectively, computed as a range of normalised frequency versus velocity (as class intervals) which will later be opened with Microsoft Excel™ or notepad. The results can be plotted as histograms to determine the characteristic velocity of fluid flow either within the joints or within the matrix.

A hydrostatic far-field fluid pressure gradient has been assigned to the models in order to give flow velocities a physical meaning. A fixed (Dirichlet) fluid pressure is applied on the two opposing edges of the model. The other two edges are treated as no flow boundaries. In the model, this makes the fractures impermeable on the no-flow boundary region.

Outputs will be velocity histograms, pressure distribution, stream lines, etc.